Course 6031A · Semester 6 · Program Elective · Revision 2021

Energy Conservation & Audit (EE)

A full professional handbook for learning energy scenario, conservation, BEE star labelling, energy management in lighting, motors and transformers, cogeneration, and energy audit methodology for electrical systems.

Course overview

This subject builds an engineer's ability to identify energy losses, select efficient electrical equipment, estimate energy savings, understand government efficiency schemes, and conduct energy audits with proper instruments and reports.

Malayalam: Energy audit □□□ energy □□□□□ □□□□□□□□□□□□, □□□□□ saving □□□□□□□, □□□ equipment efficient □□□ □□□□□□ data □□□□□□□□□□ □□□□□□□□□□□□□.

Official syllabus information

Item	Value
Program	Diploma in Electrical and Electronics Engineering
Course Code	6031A
Course Title	Energy Conservation & Audit (EE)
Semester	6
Credits	4
Category	Program Elective
Periods	4 (L:3 T:1 P:0); 60 per semester

Objectives

- Understand global energy scenarios and environmental issues.
- Apply engineering practices in energy conservation.
- Select suitable resources for sustainable development.
- Summarize Energy Conservation Act features.
- Use audit data analysis and corrective actions.

Course outcomes

- CO1: Explain conservation need and BEE labelling.
- CO2: Identify energy management techniques.
- CO3: Illustrate conservation through cogeneration.
- CO4: Summarize energy audit in electrical systems.

Learning roadmap

Study in this order: energy scenario -> equipment efficiency -> cogeneration -> audit methodology. For exam answers, always combine definition, diagram, practical benefit and example.

Redesigned syllabus roadmap

Official modules converted into a practical engineering learning route.

Module 1 · CO1 · 15 Hours

Energy Scenario, Conservation and BEE Star Labelling

Explain the need of energy conservation and BEE star labelling.

Chapter introduction

Energy conservation reduces avoidable energy waste without reducing useful work. This module connects global energy scenario, energy security, efficiency and BEE labelling.

Malayalam support: □ □□□□□□□□□□ energy conservation □□□□□□□□□□□□ □□□□□□□□□□, star labelled appliance □□□□□□ energy saving □□□□□□□□□, BEE schemes □□□□□□ efficiency improve □□□□□□□□□□ □□□□□□ □□□□□□□□□□□□.

Energy resources **Sustainable resources** **BEE schemes** **MEPS** **Star label**

Core syllabus topics

- Energy resources and classification
- Global and national energy scenario
- Energy intensity and energy efficiency
- Energy security and conservation principles
- ECBC, standards and labelling, DSM, Bachat Lamp Yojana, SME efficiency
- Comparative and endorsement labels, MEPS and star ratings

Engineering explanation

BEE star label helps compare energy consumption of appliances. A higher star rating means lower consumption for the same useful output. In audit work, star labelled devices are compared with existing loads to estimate savings.

> **Exam format:** Definition -> need -> BEE scheme -> label type -> appliance example -> saving estimate.

Expected questions

1. Explain energy conservation and its importance.
2. Explain schemes of BEE under Energy Conservation Act 2001.
3. Compare star rated and non-star rated appliances.
4. Define energy intensity and energy security.

Module 2 · CO2 · 15 Hours

Energy Management Techniques in Electrical Systems

Identify energy management techniques in lighting, motors, transformers and conservation equipment.

Lighting systems

Lighting conservation includes efficient lamps, LED retrofit, daylight use, occupancy sensors, correct lux level, timer control, and maintenance of reflectors/luminaires.

Malayalam: Lighting-□ LED, sensor, daylight use □□□□□□ □□□□□□□□□□ power consumption □□□□□□□□□□.

Motors and transformers

Energy efficient motors reduce copper, iron and mechanical losses. Correct loading, VFD use and voltage balance improve savings. Efficient transformers and amorphous transformers reduce no-load and load losses.

Conservation equipment

Soft starters reduce starting stress. Automatic star-delta converters reduce starting current. VFDs control speed for variable loads. APFC/IPFC improve power factor and reduce penalty.

In panels, APFC health, capacitor steps, contactor condition and power factor trends must be checked during audit.

Expected questions

1. Explain energy conservation techniques in lighting systems.
2. Explain features of energy efficient motors.
3. Explain amorphous transformers.
4. Draw schematic representation of VFD and APFC.

Module 3 · CO3 · 14 Hours

Energy Conservation Through Cogeneration

Cogeneration improves total efficiency by producing electricity and useful heat from the same fuel.

Need and principle

Cogeneration, or combined heat and power, uses waste heat from power generation for process heat or steam. This reduces fuel waste and improves total efficiency.

Malayalam: □□□ fuel □□□□□□□□□ electricity□□□ heat□□□ □□□□□□□□□□ waste □□□□□, efficiency □□□□□.

Types

- Steam turbine cogeneration
- Gas turbine cogeneration
- Reciprocating engine cogeneration

Prime movers

Steam turbine systems include back pressure turbine and extraction condensing turbine. Gas turbines use exhaust heat. Reciprocating engines recover jacket water and exhaust heat.

Selection criteria

Heat-to-power ratio, fuel availability, load pattern, operating hours, maintenance skill, capital cost, reliability and environmental rules decide the system.

Module 4 · CO4 · 14 Hours

Energy Audit of Electrical Systems

Energy audit uses measured data to find losses, recommend improvements and verify savings.

Audit objectives and types

Energy audit identifies where energy is used and wasted. Types are preliminary, targeted and detailed audit. Detailed audit is measurement based and gives strong recommendations.

Malayalam: Energy audit-□ measurement □□□□□□□ losses □□□□□□□□□ saving action □□□□□□□□□□□□□□□□□.

Instruments

- Power analyser
- Clamp meter
- Lux meter
- Thermal imager
- Energy meter
- Data logger

Methodology

Form team, collect bills, prepare data sheets, measure loads, analyse energy flow, identify losses, estimate savings/payback, prepare report and verify after implementation.

Sankey diagram

A Sankey diagram shows energy input, useful output and losses using proportional flow width. It is useful for explaining energy balance visually.

Formula / Rule Bank

For this subject, combine formulas, audit rules and diagram checklists.

Energy saving

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Saving = Old consumption - New consumption
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Cost saving

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Cost saving = Energy saving in kWh x Tariff per kWh
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Simple payback

Payback = Investment / Annual saving

Efficiency

Efficiency = Useful output / Input x 100%

Audit rule

Measure -> Analyse -> Recommend -> Implement -> Verify

Sankey rule

Flow width is proportional to energy quantity.

Solved Examples / Problem Lab

Step-by-step solutions suitable for exam writing.

Example 1 - Star rated appliance saving

Old appliance uses 2.4 kWh/day, star rated appliance uses 1.6 kWh/day.

Saving per day = 0.8 kWh. Annual saving = 292 kWh.

At Rs.6/kWh, cost saving = Rs.1752/year.

Example 2 - Payback

Investment = Rs.12000. Annual saving = Rs.3000.

Payback = 12000 / 3000 = 4 years.

Example 3 - Lighting retrofit

50 lamps, old 40 W, new LED 18 W, use 8 h/day.

Power saved = 1.1 kW. Daily saving = 8.8 kWh.

Monthly saving = 264 kWh.

Example 4 - Efficient motor

Input old = 12 kW, efficient motor = 10.5 kW, operation = 2000 h/year.

Annual saving = 3000 kWh.

Practice Section and Expected Questions

Module-wise exam preparation.

One mark

1. Define energy audit.
2. What is BEE?
3. Name two energy audit instruments.
4. What is a Sankey diagram?

Short answer

1. Explain need of energy conservation.
2. List advantages of star labelling.
3. Explain energy efficient motors.
4. Compare preliminary and detailed audit.

Descriptive

1. Explain BEE schemes under Energy Conservation Act.
2. Explain cogeneration systems.
3. Explain ten-step energy audit methodology.
4. Explain soft starter, VFD, APFC and IPFC.

Objective

Which diagram shows energy input, useful output and losses by proportional flow width?

Sankey diagram.

Fresh Model Question Paper

Original questions based on the syllabus.

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Time: 3 Hours **Maximum Marks:** 100

Part A

1. Define energy intensity.
2. What is MEPS?
3. Name two instruments used for energy audit.
4. What is cogeneration?
5. What is the role of Sankey diagram?

Part B

1. Explain BEE star labelling and its benefits.
2. Explain energy conservation techniques in lighting systems.
3. Explain energy efficient motors and their limitations.
4. Explain preliminary, targeted and detailed energy audits.

Part C

1. Explain schemes of BEE under Energy Conservation Act 2001.
2. Calculate annual energy and cost saving when 60 lamps of 40 W are replaced by 18 W LED lamps for 10 h/day at Rs.7/kWh.
3. Explain cogeneration systems based on steam turbine, gas turbine and reciprocating engine.
4. Explain energy audit methodology and draw a Sankey diagram.

Complete Model Answers

Use these as answer-writing guides.

Energy intensity Energy intensity is energy consumed per unit output or economic activity.

MEPSMinimum Energy Performance Standards specify minimum efficiency levels.

Lighting problem Power saved = 60 x (40-18) W = 1320 W = 1.32 kW. Daily saving = 13.2 kWh. Annual saving = 4818 kWh. Cost saving = Rs.33726.

Energy audit methodology Form team, collect utility bills, conduct walk-through survey, measure loads, analyse energy flow, identify losses, estimate savings/payback, prepare report, implement and verify.